

Lithium Falsification Criteria

FFGFT predictions for light element
abundances

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May 2026

Preamble

This document formulates the falsification criterion for the FFGFT position on the cosmological lithium problem, in direct response to P. Austin’s methodological question: *“What abundance pattern would distinguish the FFGFT equilibrium interpretation from BBN plus stellar depletion?”*

Lithium is the most difficult of the three falsification tests. Casimir (Doc. 220) and redshift (Doc. 221) build on local ξ_0 -vacuum action; lithium tests the *cosmic pre-history* of the universe. The FFGFT answer here is not a modification of a single reaction rate, but a fundamentally different view on the origin of the light elements.

1 The cosmological lithium problem

1.1 Observational situation

The principal light-element abundances in metal-poor population II stars (Spite plateau, $[\text{Fe}/\text{H}] < -2$):

Element	Observed	BBN prediction	Status
⁴ He	24.8 % mass fraction	24.7-24.8 %	✓
D/H	$(2.5 \pm 0.2) \cdot 10^{-5}$	$(2.5 \pm 0.1) \cdot 10^{-5}$	✓
⁷ Li/H	$(1.6 \pm 0.3) \cdot 10^{-10}$	$(5.0 \pm 0.7) \cdot 10^{-10}$	factor 3
⁶ Li/H	$\lesssim 10^{-12}$	$\lesssim 10^{-14}$	marginal

While ⁴He and D/H are consistent with BBN+Planck, ⁷Li is short by a factor of 3-4. This discrepancy has been unresolved for two decades.

1.2 Three standard solution attempts

- 1. **Modified reaction rates:** CERN n_TOF experiments on ⁷Be destruction rates. So far no sufficient correction found.
- 2. **Stellar depletion:** mixing, diffusion, atmospheric depletion in Spite stars. Consistent with ⁷Li loss, but the Spite plateau is too smooth for depletion models.
- 3. **Structure formation effects:** Pop-III stellar capture (Aanda 2025, Yeh & Olive 2022). Explains part of the discrepancy, depends on Pop-III IMF.

2 FFGFT position: no BBN phase

2.1 Static universe vs. standard BBN

Standard BBN presupposes a *cooling, expanding phase* from $T \sim 10^{11}$ K to $T \sim 10^9$ K in ~ 3 minutes. In this phase:

- the deuterium bottleneck at $T \sim 10^9$ K is the rate-limiting step,

- the destruction of ${}^7\text{Be}$ by electron capture ($t_{1/2} \approx 53$ days) is too slow to complete before *cosmological cooldown*,
- the resulting ${}^7\text{Li}$ level is a *frozen-in remnant* of an interrupted reaction sequence.

In FFGFT this phase does not exist. The universe is static; there is no cosmological cooldown from 10^{11} to 10^9 K in 3 minutes. The cosmic microwave background has always been at $T_{\text{CMB}} = 2.725$ K (see Doc. 091, $\rho_{\text{CMB}} = \xi_0 \hbar c / L_\xi^4$). At this temperature all atomic nuclei are stable, and there are no BBN reactions.

Consequence: the observed abundances of ${}^4\text{He}$, D, ${}^7\text{Li}$, ${}^6\text{Li}$ in metal-poor stars are in FFGFT *not* primordial in the standard sense. They result from a *stellar prehistory* with unbounded time.

2.2 Stellar sources of light elements

In the FFGFT view, the abundances arise from the long-term equilibrium between astrophysical production and destruction mechanisms:

- **${}^4\text{He}$:** primary producer is hydrogen fusion in stars of all masses. In a universe with unbounded cosmic time, Pop-III stars and successive generations enrich the interstellar medium to $\sim 25\%$ He. Consistent with present observation.
- **D:** D is *consumed* by stellar fusion and *produced* by cosmic spallation on helium nuclei. Equilibrium gives $D/H \sim 10^{-5}$ – consistent with observation.
- **${}^7\text{Li}$:** produced in AGB stars, classical novae, cosmic spallation. Destroyed in stellar surfaces by p-capture. Equilibrium level $\sim 10^{-10}$ – consistent with the Spite plateau.
- **${}^6\text{Li}$:** almost exclusively from cosmic spallation; no BBN contribution expected. Level $\lesssim 10^{-12}$ consistent with sparse upper bounds.

What this changes: the “lithium problem” disappears in this view, because there is no BBN prediction value exceeding the observation. Instead, the task is to show that stellar sources quantitatively reproduce the Spite plateau level – which is exactly what the Aanda 2025 paper and Pop-III modelling are pursuing, albeit within the expansion paradigm.

3 FFGFT predictions

3.1 Constancy of ξ_0 over cosmic time

The *most important* FFGFT prediction in the lithium domain is that $\xi_0 = 4/30000$ **is a time-independent geometric constant**. Hence:

- the fine-structure constant $\alpha = \xi_0 \cdot E_0^2$ is constant (see Doc. 202, §22),
- all nuclear binding energies (Coulomb part $\propto \alpha$, nuclear interaction scale-free) are constant,
- all reaction rates (cosmic spallation, stellar nucleosynthesis) are constant.

Falsification: variation tests of natural constants (quasar spectrum analyses across large z , Webb & Murphy et al., many-multiplet method) show no variation in most studies; an unambiguous detection $|\Delta\alpha/\alpha| > 10^{-5}$ between $z = 0$ and $z = 4$ would refute FFGFT. *Currently:* maximum $|\Delta\alpha/\alpha| < 10^{-6}$, FFGFT consistent.

3.2 Constancy of element abundances in old structures

In standard BBN+expansion, all abundances are $\propto \eta$ (baryon-to-photon ratio) and thus time-dependent in the early phase. In FFGFT, η is a position-dependent quantity of the present cosmic matter distribution, not a time-evolving parameter.

FFGFT prediction: element abundances in old, metal-poor stars show a *smooth plateau* without systematic dependence on the redshift of observation. In the standard BBN+expansion picture, slight variations would be expected at very high redshifts ($z > 6$), since these observations probe closer to the actual nucleosynthesis.

Falsification: if JWST observations of galaxies at $z > 6$ systematically find *different* abundances than the local Spite plateau, the FFGFT view (abundance as stellar equilibrium, not primordial value) is challenged.

3.3 Quantitative prediction: no ${}^7\text{Li}$ discrepancy

In FFGFT the factor of 3 between BBN prediction and observation *vanishes* – not through a modified reaction rate, but because the BBN prediction is not the correct reference. The correct reference is the *stellar equilibrium*, lying at ${}^7\text{Li}/\text{H} \sim 10^{-10}$.

Falsification: if the standard BBN prediction can be reduced to the observed value by corrections to nuclear reaction rates (e.g. new ${}^7\text{Be}(n, \alpha){}^4\text{He}$ resonances from n_TOF) *without* worsening D/H and ${}^4\text{He}$, the FFGFT position becomes superfluous (but not refuted). A genuine refutation would be a *complete, consistent* standard BBN solution for all four abundances without additional assumptions.

3.4 ${}^4\text{He}$ abundance as critical test

The *hardest* internal test of the FFGFT position is the ${}^4\text{He}$ abundance of 24.8% in extremely metal-poor stars. Standard BBN explains this number in the first 3 minutes of cosmic history. FFGFT must explain this number through stellar prehistory.

FFGFT requirement: Pop-III stars (or comparable early populations) must produce sufficient He enrichment to reach $\sim 25\%$ mass fraction, without destroying ${}^7\text{Li}$, D, or metal abundances in Spite stars. The available cosmic

time in FFGFT is *unbounded*, which opens this path – but it must be quantitatively consistent.

Falsification: a Pop-III/II stellar model that *provably* delivers either too little or too much ^4He while simultaneously reproducing the other three abundances.

4 Specific falsification criteria

FFGFT is **refuted** by the following observations:

1. **Variation of α or ξ_0 :** a significant variation $|\Delta\alpha/\alpha| > 10^{-5}$ between $z = 0$ and high redshifts. *Current status:* no variation detected, FFGFT consistent.
2. **Abundance variation with z :** if JWST spectroscopy of light elements in galaxies at $z > 6$ systematically finds different abundances than the local Spite plateau, not explainable by local chemical evolution. *Current status:* undecided, critical test in coming years.
3. **Complete standard BBN solution:** a consistent modification of nuclear reaction rates (e.g. from n_{TOF}) that reproduces all four abundances (^4He , D, ^7Li , ^6Li) *without* additional free parameters within the standard expansion framework. This would not refute FFGFT geometry, but would render the FFGFT position on the lithium problem superfluous.
4. **Inconsistency of stellar sources:** a quantitative stellar model showing that observed abundances cannot be reproduced from stellar sources + cosmic spallation without invoking BBN. *Current status:* Aanda 2025 work shows that Pop-III modelling can explain the Spite plateau. Full consistency for all four abundances without BBN is not yet demonstrated.
5. **^4He variation in old structures:** if ^4He abundance in extremely metal-poor high-redshift HII regions systematically deviates from 24.8%. *Current status:* cur-

rent measurements at moderate redshifts all give $\sim 24.8\%$.

5 What FFGFT does NOT predict

- FFGFT does *not* predict a specific numerical value for ${}^7\text{Li}/\text{H} = X$ competing with BBN. The position is methodological: BBN is not the reference, because there is no BBN phase.
- FFGFT does *not* predict any new exotic reaction converting ${}^7\text{Li}$. The solution lies in the *history*, not in the *physics of a single reaction*.
- FFGFT does *not* predict any anomaly of stellar fusion or spallation rates. Both processes proceed with standard cross sections.

6 Connection to Doc. 220 and 221

The three falsification documents form a hierarchical structure:

- **Doc. 220 (Casimir):** tests ξ_0 -vacuum on $\sim 100\,\mu\text{m}$ scale. Local, directly measurable.
- **Doc. 221 (Redshift):** tests the same ξ_0 mechanism on cosmological scales. Astronomical, indirect but precise.
- **Doc. 222 (Lithium):** tests the *cosmic history* – that the universe is static and ξ_0 is constant over cosmic time. Model-based, since direct temporal observation of element synthesis is impossible.

Hierarchy of strength:

- Casimir falsification (Doc. 220) is the *strongest* – immediately measurable, independent of model assumptions.
- Redshift falsification (Doc. 221) is *strong* – the H_0 prediction is parameter-free, several independent tests available.

- Lithium falsification (this document) is the *weakest* – because it builds on astrophysical models that themselves carry assumptions. It is more a *consistency argument* than a sharp falsification test.

Consequence for experimental strategy: those wishing to test FFGFT should begin with Casimir measurements in the 30–100 μm range (Doc. 220), then with expansion-free H_0 measurements (GW standard sirens, Doc. 221), and only afterwards – when the first two yield positive indications – with cosmological abundance consistency analyses (this document).

7 Summary

FFGFT positions itself on the cosmological lithium problem **not** with a modified BBN reaction rate, but with a fundamentally different view: in a static universe with time-independent ξ_0 , there is no BBN phase, and the observed light element abundances result from an unbounded stellar prehistory.

The FFGFT predictions:

- α and ξ_0 are constant over cosmic time.
- Element abundances show no systematic z -profile beyond what is explained by local chemical evolution.
- Pop-III stellar models reproduce ${}^4\text{He}$, D, ${}^7\text{Li}$, ${}^6\text{Li}$ with a consistent set of astrophysical sources and sinks.

Methodological note. This position is FFGFT-internal. It follows from the static universe (Doc. 025), the ξ_0 energy loss for redshift (Doc. 064/221), and the time-independent geometric constant (Doc. 202).

Status. The FFGFT position is compatible with all current observations. It is tested by the combination of α variation tests (negative, FFGFT-consistent), z -abundance profiles (open, JWST-critical), and Pop-III consistency modelling (in progress, Aanda 2025). An unambiguous refu-

tation requires several independent observational pillars all pointing against FFGFT – individual anomalies can be explained consistently.